

COMPLEX NUMBERS

ENSC 180 – Introduction to Engineering Analysis Tools

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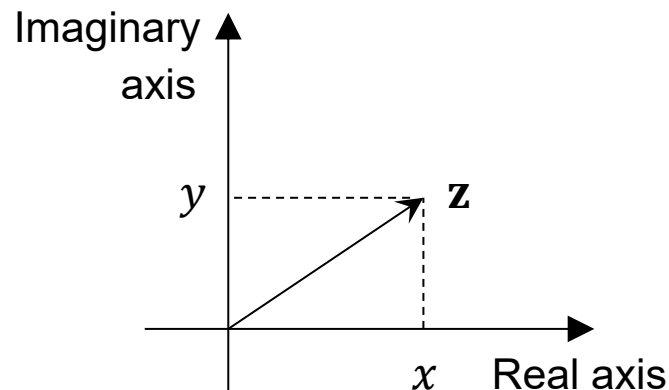


COMPLEX NUMBERS

- Complex numbers have the form $z = x + iy$, where $i^2 = -1$ (that is, $i = \sqrt{-1}$); Sometimes j used instead of i , imaginary unit
- This representation is referred to as the **rectangular form**
- x and y are real numbers, referred to, respectively, as the ***real*** and ***imaginary parts*** of the complex number:

$$x = \text{Re}(z), \quad y = \text{Im}(z)$$

- The complex number $z = x + iy$ can also be represented as a vector $\mathbf{z} = (x, y)$ in a 2D plane whose axes are called real and imaginary axis



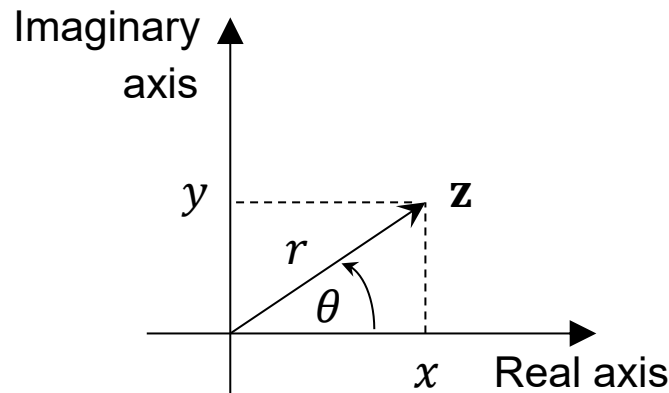
COMPLEX NUMBERS IN POLAR FORM

- Complex numbers can also be represented in **polar form**
- The polar form shows complex number's **magnitude** and **phase**:

$$r = |\mathbf{z}| = \sqrt{x^2 + y^2}, \quad \theta = \angle \mathbf{z} = \arctan \frac{y}{x} \quad (+\pi, \text{ sometimes})$$

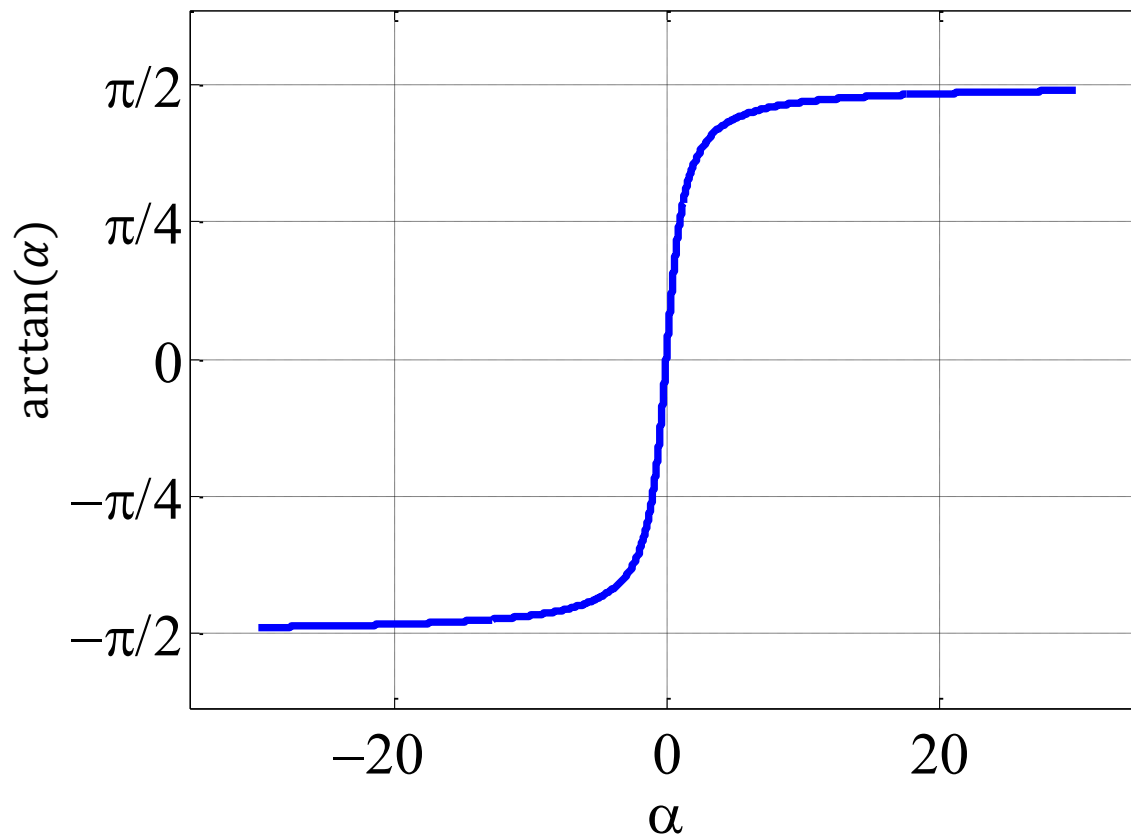
- Going from polar to rectangular form is also easy:

$$x = r \cdot \cos \theta, \quad y = r \cdot \sin \theta$$



PHASE OF A COMPLEX NUMBER

Plot of $\arctan(\alpha)$ for $-30 < \alpha < 30$

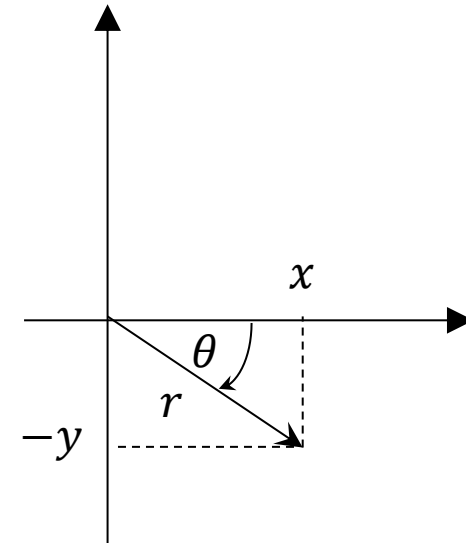
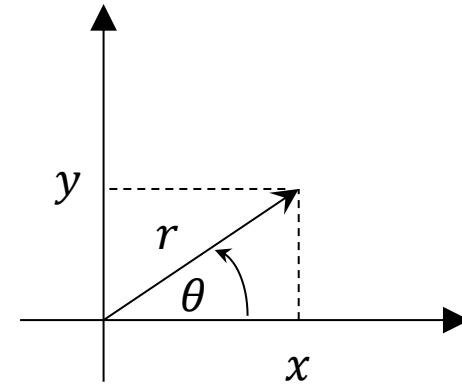


PHASE OF A COMPLEX NUMBER

- Let $x, y > 0$
- When the complex number z is in the first quadrant ($Re(z) > 0, Im(z) > 0$)
 $\cos \theta = \frac{x}{r}, \sin \theta = \frac{y}{r}, \tan \theta = \frac{y}{x}$
 $\theta = \arctan \frac{y}{x} \in \left(0, \frac{\pi}{2}\right)$
- When the complex number z is in the fourth quadrant ($Re(z) > 0, Im(z) < 0$)

$$\cos \theta = \frac{x}{r}, \sin \theta = \frac{-y}{r}, \tan \theta = \frac{-y}{x}$$

$$\theta = \arctan \frac{-y}{x} = -\arctan \frac{y}{x} \in \left(-\frac{\pi}{2}, 0\right)$$



PHASE OF A COMPLEX NUMBER

- When the complex number z is in the second quadrant ($\operatorname{Re}(z) < 0, \operatorname{Im}(z) > 0$)

$$\cos \theta = \frac{-x}{r}, \sin \theta = \frac{y}{r}, \tan \theta = \frac{y}{-x} = \frac{-y}{x}$$

If using arctan formula, we get

$$\theta = \arctan \frac{y}{-x} = \arctan \frac{-y}{x} \in \left(-\frac{\pi}{2}, 0\right)$$

as for the fourth quadrant - incorrect!

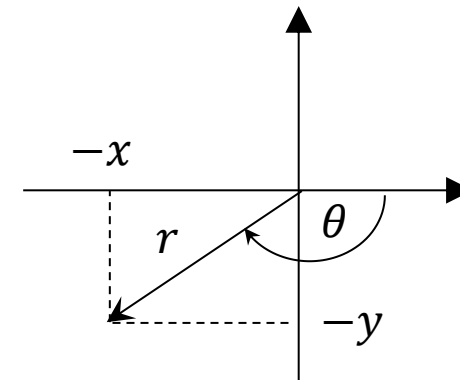
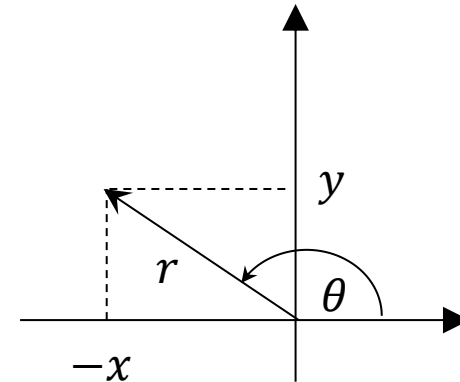
The correct formula is

$$\theta = \arctan \frac{y}{-x} + \pi \in \left(\frac{\pi}{2}, \pi\right)$$

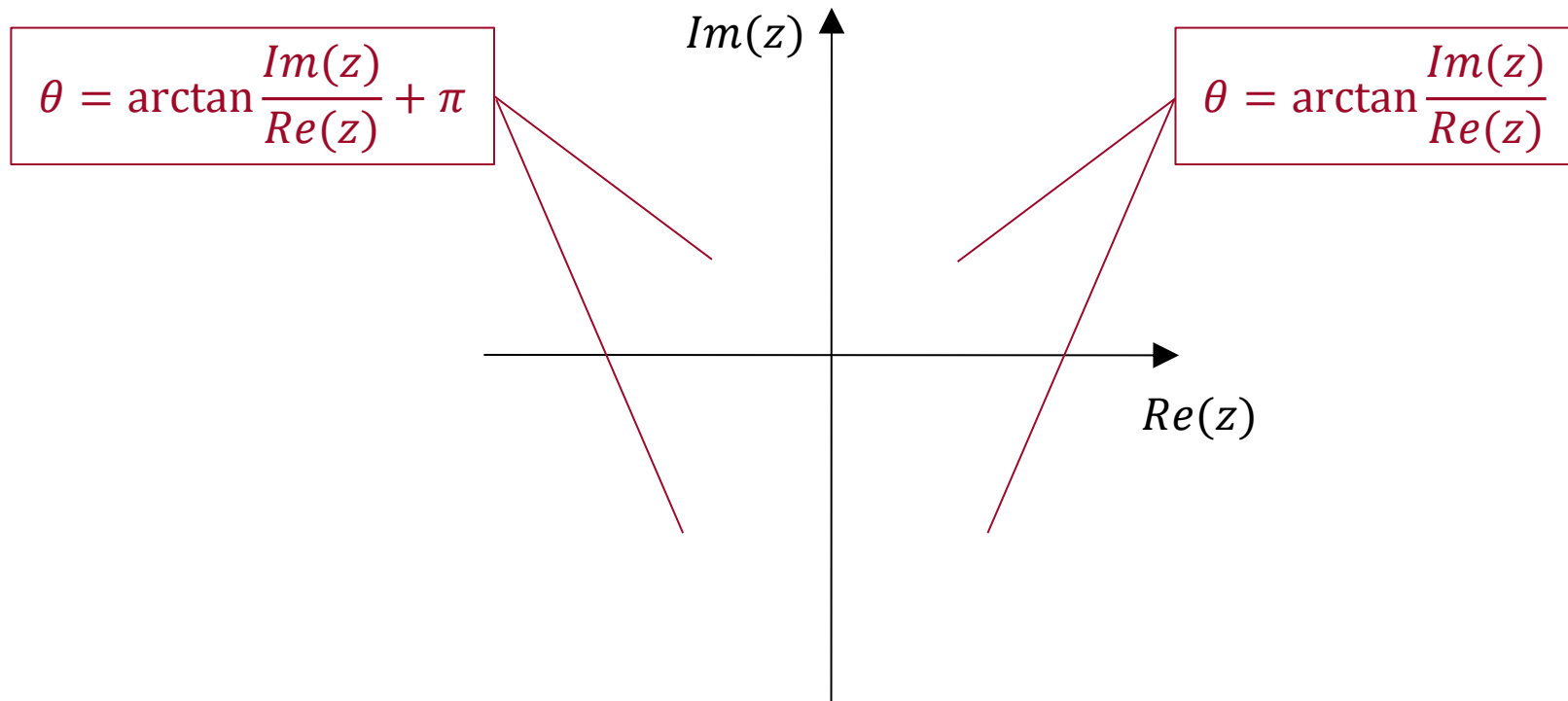
- When the complex number z is in the third quadrant ($\operatorname{Re}(z) < 0, \operatorname{Im}(z) < 0$)

$$\cos \theta = \frac{-x}{r}, \sin \theta = \frac{-y}{r}, \tan \theta = \frac{-y}{-x} = \frac{y}{x}$$

Again, the correct formula is $\theta = \arctan \frac{-y}{-x} + \pi \in \left(\pi, \frac{3\pi}{2}\right) \equiv \left(-\pi, -\frac{\pi}{2}\right)$



PHASE OF A COMPLEX NUMBER



PHASE OF A COMPLEX NUMBER

- In MATLAB, check the following

```
>> z = -3+j*2
```

```
z =
```

```
-3.0000 + 2.0000i
```

```
>> atan(2/(-3))
```

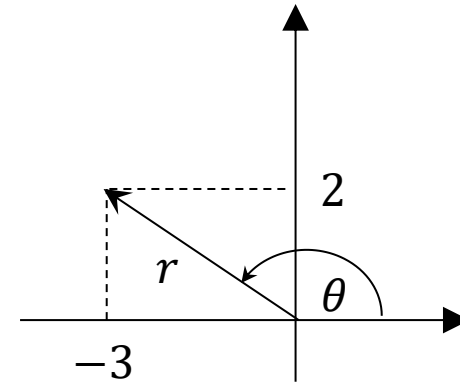
```
ans =
```

```
-0.5880
```

```
>> angle(z)
```

```
ans =
```

```
2.5536
```



- Using `atan` (which is `arctan` in MATLAB) by itself gives incorrect result, as we expected; adding `pi` would make it correct
- MATLAB function `angle` gives the correct result

EULER'S FORMULA

- **Euler's formula**, named after Swiss mathematician Leonhard Euler, relates $e^{i\theta}$ function to trigonometric functions:

$$e^{i\theta} = \cos\theta + i \cdot \sin\theta$$

- This relationship provides another way to express the polar form of the complex number

$$\begin{aligned} z &= x + iy \\ &= r\cos\theta + i \cdot r\sin\theta \\ &= r(\cos\theta + i \cdot \sin\theta) \\ &= re^{i\theta} \end{aligned}$$

- Therefore, in engineering, it is common to write the polar form of the complex number as

$$z = re^{i\theta} \equiv r\angle\theta$$

COMPLEX NUMBERS IN MATLAB

- MATLAB allows you to define a complex number in either rectangular or polar form
- Rectangular form :

```
>> z = 1+2*i
```

```
>> z = 1+2*j
```

```
>> z = 1+2i
```

```
>> z = 1+2j
```

All of these will define
 $z = 1 + 2i$

- Polar form:

```
>> z = 2*exp(3*i)
```

```
>> z = 2*exp(3*j)
```

```
>> z = 2*exp(3i)
```

```
>> z = 2*exp(3j)
```

All of these will define
 $z = 2e^{j3} = -1.980 + 0.282i$

EXAMPLE: RECTANGULAR TO POLAR CONVERSION

- Convert $z = 4 + 3i$ to polar form in MATLAB

```
>> z = 4+3i
```

```
z =
```

```
4.0000 + 3.0000i
```

```
>> r = abs(z)
```

```
r =
```

```
5
```

```
>> theta_rad = angle(z)
```

```
theta_rad =
```

```
0.6435
```

```
>> theta_deg = theta_rad*180/pi
```

```
theta_deg =
```

```
36.8699
```

$$r = \sqrt{4^2 + 3^2} = 5$$

$$\theta = \arctan \frac{3}{4} = 0.6435 \text{ rad}$$

$$0.6435 \text{ rad} = 36.8699^\circ$$

EXAMPLE: POLAR TO RECTANGULAR CONVERSION

- Convert $z = 4e^{i \cdot 2}$ to rectangular form in MATLAB

```
>> z = 4*exp(2*i)
```

```
z =
```

```
-1.6646 + 3.6372i
```

- By default, MATLAB will show the rectangular representation
- We can also get real and imaginary parts using MATLAB functions `real` and `imag`

```
>> real(z)
```

```
ans =
```

```
-1.6646
```

$$\begin{aligned} \operatorname{Re}(z) &= r \cos \theta = 4 \cdot \cos 2 \\ &= -1.6646 \end{aligned}$$

```
>> imag(z)
```

```
ans =
```

```
3.6372
```

$$\begin{aligned} \operatorname{Im}(z) &= r \sin \theta = 4 \cdot \sin 2 \\ &= 3.6372 \end{aligned}$$

COMPLEX CONJUGATE

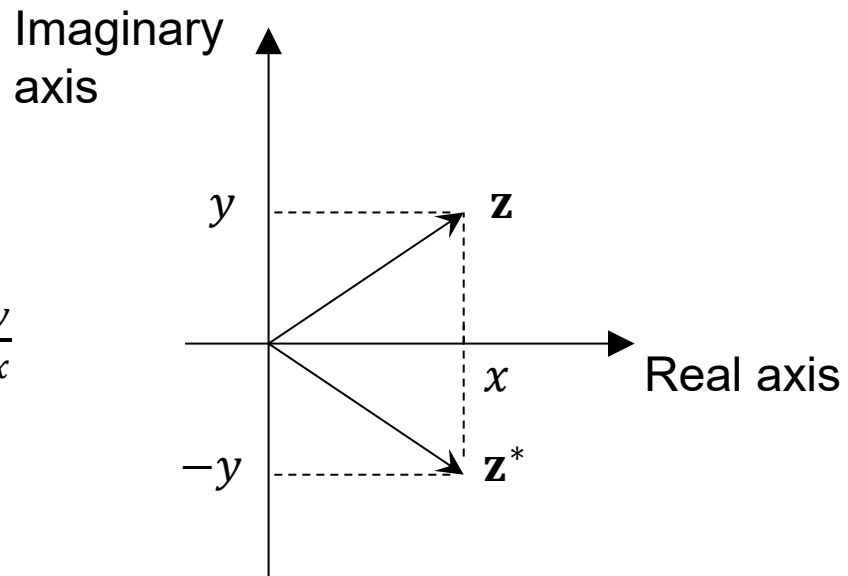
- The **complex conjugate** of $z = x + iy$ is $z^* = x - iy$
- In polar form, if $z = re^{i\theta}$, then $z^* = re^{-i\theta}$
- Complex conjugation means changing the sign of the imaginary part of the complex number (in rectangular form), or changing the sign of phase (in polar form)

$$\operatorname{Re}(z^*) = \operatorname{Re}(z) = x$$

$$\operatorname{Im}(z^*) = -\operatorname{Im}(z) = -y$$

$$|z^*| = |z| = r = \sqrt{x^2 + y^2}$$

$$\angle z^* = -\angle z = -\theta = -\arctan \frac{y}{x}$$



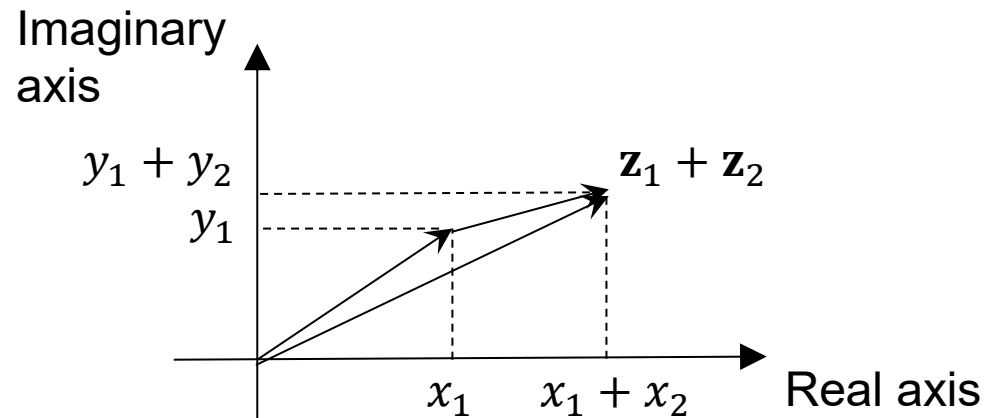
- MATLAB function: `conj`

ADDITION OF COMPLEX NUMBERS

- Complex addition – add real and imaginary parts separately
- If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$z_1 + z_2 = (x_1 + x_2) + i \cdot (y_1 + y_2)$$

- This can also be represented as vector addition



MULTIPLICATION OF COMPLEX NUMBERS

- Complex multiplication is like multiplying sums of real numbers, but remembering that $i^2 = -1$
- If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= x_1 x_2 + ix_1 y_2 + iy_1 x_2 + i^2 y_1 y_2 \\ &= x_1 x_2 - y_1 y_2 + i \cdot (x_1 y_2 + y_1 x_2) \end{aligned}$$

- Complex multiplication is easier in polar form, using the standard rules for multiplying exponentials
- If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then

$$z_1 z_2 = r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

MULTIPLICATION BY COMPLEX CONJUGATE

- The product of a complex number and its complex conjugate is its magnitude
- Let $z = x + iy$, then

$$\begin{aligned} zz^* &= (x + iy)(x - iy) \\ &= x^2 + y^2 + i \cdot (yx - xy) \\ &= x^2 + y^2 \\ &= |z|^2 \end{aligned}$$

- This is easier to see in the polar form
- If $z = re^{i\theta}$ then

$$zz^* = re^{i\theta}re^{-i\theta} = r^2e^{i(\theta-\theta)} = r^2e^{i0} = r^2 = |z|^2$$

DIVISION OF COMPLEX NUMBERS

- Division of complex numbers can be cumbersome in rectangular form
- If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{z_1 z_2^*}{z_2 z_2^*} \\ &= \frac{x_1 x_2 + y_1 y_2 + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \\ &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i \cdot \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}\end{aligned}$$

- It is much easier in polar form, using division of exponentials
- If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

APPLICATIONS OF COMPLEX NUMBERS

- Complex numbers arise naturally in finding the roots of polynomials
- For example, the solutions of

$$z^2 + 1 = 0$$

are

$$z = i \quad \text{and} \quad z = -i$$

- In MATLAB

```
>> roots([1 0 1])
```

```
ans =
```

```
0.0000 + 1.0000i
```

```
0.0000 - 1.0000i
```

Polynomial coefficients
expressed as a vector

$$1 \cdot z^2 + 0 \cdot z + 1 \rightarrow [1 \ 0 \ 1]$$

APPLICATIONS OF COMPLEX NUMBERS

- Some algebraic or calculus problems can be solved more easily using complex numbers and complex arithmetic
- For example, using real-number arithmetic, we know that

$$(\cos x + i \sin x)^n = \sum_{k=0}^n \binom{n}{k} (\cos x)^k (i \sin x)^{n-k}$$

but it is not clear how to simplify it further.

- Using complex arithmetic, however, we can easily conclude that

$$(\cos x + i \sin x)^n = (e^{ix})^n = e^{inx} = \cos nx + i \sin nx$$

APPLICATIONS OF COMPLEX NUMBERS

- Another example, this time from calculus

$$\int \sin^2 x \cos 4x \, dx = ?$$

- This integral may be easier to compute using complex arithmetic.
- First, recall Euler's formula $e^{ix} = \cos x + i \cdot \sin x$. From this, it is easy to see that

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix}) \quad \text{and} \quad \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

- Using this in the integral above, we obtain

$$\begin{aligned} \int \sin^2 x \cos 4x \, dx &= \int \left\{ \frac{1}{2i}(e^{ix} - e^{-ix}) \right\}^2 \left\{ \frac{1}{2}(e^{i4x} + e^{-i4x}) \right\} dx \\ &= -\frac{1}{8} \int (e^{i2x} - 2 + e^{-i2x})(e^{i4x} + e^{-i4x}) dx \end{aligned}$$

APPLICATIONS OF COMPLEX NUMBERS

$$\begin{aligned}\int \sin^2 x \cos 4x \, dx &= \int \left\{ \frac{1}{2i} (e^{ix} - e^{-ix}) \right\}^2 \left\{ \frac{1}{2} (e^{i4x} + e^{-i4x}) \right\} dx \\&= -\frac{1}{8} \int (e^{i2x} - 2 + e^{-i2x})(e^{i4x} + e^{-i4x}) dx \\&= -\frac{1}{8} \int (e^{i6x} - 2e^{i4x} + e^{i2x} + e^{-i2x} - 2e^{-i4x} + e^{-i6x}) dx \\&= -\frac{1}{8} \left\{ \frac{e^{i6x}}{i6} - \frac{2e^{i4x}}{i4} + \frac{e^{i2x}}{i2} + \frac{e^{-i2x}}{-i2} - \frac{2e^{-i4x}}{-i4} + \frac{e^{-i6x}}{-i6} \right\} + C \\&= -\frac{1}{8} \left\{ \sin 2x - \sin 4x + \frac{1}{3} \sin 6x \right\} + C\end{aligned}$$

APPLICATIONS OF COMPLEX NUMBERS

- In circuit theory, electronics, and some areas of physics, complex numbers are used to represent sinusoidal signals (usually voltages)
- Such representation is called a **phase vector**, or **phasor**
- For example, in circuit analysis we often have to add AC voltages
- Consider two AC voltages with the same frequency f , but different amplitudes and phases

$$v_1(t) = A_1 \cos(2\pi ft + \theta_1)$$

$$v_2(t) = A_2 \cos(2\pi ft + \theta_2)$$

- Using trigonometric identities, we can show that [from Wikipedia]

$$v_1(t) + v_2(t) = A_3 \cos(2\pi ft + \theta_3)$$

where

$$A_3 = \sqrt{(A_1 \cos \theta_1 + A_2 \cos \theta_2)^2 + (A_1 \sin \theta_1 + A_2 \sin \theta_2)^2}$$

$$\theta_3 = \arctan \left(\frac{A_1 \sin \theta_1 + A_2 \sin \theta_2}{A_1 \cos \theta_1 + A_2 \cos \theta_2} \right)$$

APPLICATIONS OF COMPLEX NUMBERS

- This is much easier using complex numbers
- Note that

$$v_1(t) = A_1 \cos(2\pi ft + \theta_1) = \operatorname{Re}(A_1 e^{i(2\pi ft + \theta_1)}) = \operatorname{Re}(A_1 e^{i\theta_1} e^{i2\pi ft})$$

$$v_2(t) = A_2 \cos(2\pi ft + \theta_2) = \operatorname{Re}(A_2 e^{i(2\pi ft + \theta_2)}) = \operatorname{Re}(A_2 e^{i\theta_2} e^{i2\pi ft})$$

- We will represent the two voltages by complex numbers (phasors) whose real parts are equal to the voltages

$$v_1(t) \rightarrow A_1 e^{i\theta_1} = A_1 \angle \theta_1$$

$$v_2(t) \rightarrow A_2 e^{i\theta_2} = A_2 \angle \theta_2$$

- Note that the factors $e^{i2\pi ft}$ have been removed from the phasors, since they are common to both

APPLICATIONS OF COMPLEX NUMBERS

- To add two AC voltages of the same frequency, we can
 1. Represent each voltage as a complex number (phasor)
 2. Add the complex numbers
 3. Take the magnitude and phase of the result
- Example: $v_1(t) = 5 \cos(2\pi 60t + 25^\circ)$, $v_2(t) = 7 \cos(2\pi 60t + 42^\circ)$,
- In phasor representation: $v_1(t) = 5\angle 25^\circ$, $v_2(t) = 7\angle 42^\circ$

```
>> v1 = 5*exp(i*25*pi/180)
```

```
v1 = 4.5315 + 2.1131i
```

```
>> v2 = 7*exp(i*42*pi/180)
```

```
v2 = 5.2020 + 4.6839i
```

```
>> abs(v1+v2)
```

```
ans = 11.8719
```

```
>> angle(v1+v2)*180/pi
```

```
ans = 34.9269
```

$$5\angle 25^\circ + 7\angle 42^\circ = 11.9\angle 34.9^\circ$$

$$v_1(t) + v_2(t) = 11.9 \cos(2\pi 60t + 34.9^\circ)$$

APPLICATIONS OF COMPLEX NUMBERS

Illustration from Wikipedia

<http://en.wikipedia.org/wiki/File:Sumafasores.gif>

